

SAIGO, Japan

WMO: 477400

Lat: 36.2044N

Lon: 133.3339E

Elev: 28

StdP: 100.99

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	-0.9	-7.2	2.0	2.0	-6.0	2.3	2.8	10.9	4.7	9.8	5.4	3.1	310	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	31.5	25.6	30.5	25.3	29.5	24.9	26.5	29.9	25.9	29.3	25.4	28.7	4.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.5	20.7	28.7	24.9	20.0	28.3	24.3	19.4	27.8	82.8	29.9	80.4	29.4	78.2	28.8	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.4	7.4	-3.7	33.2	1.4	1.1	-4.7	33.9	-5.5	34.6	-6.2	35.2	-7.2	35.9		
			-4.7	27.3	1.3	0.8	-5.6	27.9	-6.4	28.4	-7.1	28.9	-8.1	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	4.3	4.5	7.0	11.8	16.5	20.4	24.9	22.3	17.0	11.8	7.0		
		DBStd	8.03	2.49	2.71	3.13	3.23	2.51	2.24	2.46	2.08	2.55	2.71	2.99		
		HDD10.0	568	177	156	103	17	0	0	0	0	0	15	100		
		HDD18.3	2046	434	388	351	197	66	6	0	1	57	195	351		
		CDD10.0	2235	1	2	11	71	203	313	461	509	370	217	71		
		CDD18.3	670	0	0	0	1	11	68	203	251	121	16	0		
		CDH23.3	4872	0	0	0	2	37	212	1538	2420	634	29	0		
		CDH26.7	1281	0	0	0	0	2	18	378	776	105	2	0		
(14)	Wind	WSAvg	3.3	3.6	3.4	3.6	3.5	3.2	2.9	3.2	3.1	3.1	3.5	3.7		
(15)	Precipitation	PrecAvg	1826	169	111	125	118	141	165	211	142	223	119	138		
(16)		PrecMax	2402	310	195	197	246	282	452	556	387	392	253	218		
(17)		PrecMin	1353	96	61	46	25	55	34	3	20	40	51	68		
(18)		PrecStd	217	48	34	37	54	69	95	130	104	100	56	44		
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	13.8	15.7	18.6	23.0	26.1	28.7	32.3	33.1	30.7	25.7	21.4	16.7		
(20)		0.4% MCWB	10.9	12.2	12.7	15.1	17.7	21.8	25.5	25.9	25.1	21.5	17.1	13.3		
(21)		2% DB	11.7	13.1	16.1	20.7	24.3	26.8	31.1	32.0	29.1	24.0	19.7	14.9		
(22)		2% MCWB	8.2	10.1	11.4	14.3	17.4	21.3	25.3	25.8	24.4	20.1	15.9	11.4		
(23)		5% DB	10.3	11.2	14.6	19.2	22.9	25.6	29.9	31.1	27.8	22.9	18.7	13.5		
(24)		5% MCWB	6.9	8.1	10.8	13.9	16.6	21.1	24.9	25.6	23.6	18.7	15.0	9.9		
(25)		10% DB	9.0	9.7	13.1	17.7	21.7	24.4	28.8	30.2	26.5	21.8	17.5	12.3		
(26)		10% MCWB	5.9	6.7	9.7	13.2	16.2	20.6	24.5	25.4	22.8	17.9	14.0	8.9		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	11.2	13.0	14.0	17.8	20.5	23.8	26.6	27.5	26.0	22.7	18.2	14.5		
(28)		0.4% MCDB	13.1	15.3	16.4	20.2	22.7	26.6	30.2	31.0	29.5	25.0	19.9	15.7		
(29)		2% WB	9.0	10.5	12.5	15.9	19.1	23.0	25.9	26.7	25.0	21.2	16.8	12.0		
(30)		2% MCDB	11.3	12.4	15.0	18.9	21.7	25.3	29.5	30.2	28.1	23.3	19.1	14.2		
(31)		5% WB	7.4	8.6	11.4	14.8	18.2	22.1	25.4	26.3	24.3	19.8	15.7	10.6		
(32)		5% MCDB	9.6	10.7	14.0	17.8	21.3	24.5	28.8	29.7	27.1	21.8	18.0	13.0		
(33)		10% WB	6.2	7.1	10.1	13.8	17.3	21.3	24.9	25.9	23.5	18.6	14.4	9.2		
(34)		10% MCDB	8.7	9.5	12.8	16.8	20.5	23.6	28.1	29.2	26.0	21.1	17.0	11.8		
(35)	Mean Daily Temperature Range	MDBR	6.1	7.0	8.3	8.7	8.1	6.5	5.4	6.2	6.5	7.7	7.5	6.5		
(36)		5% DB MCDBR	8.7	9.7	10.6	10.8	10.5	8.2	7.3	7.4	7.4	8.3	8.6	8.4		
(37)		5% WB MCWBR	6.4	7.2	7.4	6.3	5.7	4.2	3.2	3.2	3.9	5.0	5.8	6.1		
(38)		5% WB MCDBR	7.9	9.0	9.4	8.6	8.1	5.9	6.6	6.6	6.3	6.7	7.6	7.9		
(39)	Clear-Sky Solar Irradiance	taub	0.448	0.480	0.548	0.562	0.561	0.576	0.575	0.527	0.484	0.459	0.441	0.442		
(40)		taud	1.982	1.920	1.789	1.810	1.871	1.917	1.983	2.130	2.188	2.126	2.087	2.004		
(41)		Ebn at Noon	700	732	720	740	749	736	733	762	773	749	707	674		
(42)		Edh at Noon	138	165	205	211	201	192	179	152	136	132	123	127		
(44)	All-Sky Solar Radiation	RadAvg	1.68	2.48	3.73	4.88	5.31	5.12	5.00	5.21	4.03	3.25	2.14	1.50		
(45)		RadStd	0.14	0.22	0.31	0.42	0.52	0.38	0.67	0.61	0.32	0.28	0.18	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.31	N/A	N/A	N/A	N/A

Nomenclature: See separate page